THE MARITIME SAFETY COMMITTEE, RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee, RECALLING ALSO resolution A.886(21), by which the Assembly resolved that the function of adopting performance standards and tec hnical specifications, as well as amendments thereto, shall be performed by the Maritime Safety Committee and/or the Marine Environment Protection Committee, as appropriate, on behalf of the Organization, RECALLING FURTHER regulations V/19 and V/27 of the International Convention for the --- RECALLING FURTHER regulations V/19 and V/27 of the International Convention for the Safety of Life at Sea (SOLAS), 1974, which requires all ships to carry adequate and up-to-date charts, sailing directions, lists of lights, notices to mariners, tide tables and all other nautical publications necessary for the intended voyage, RECALLING resolutions A.817(19), as amended, and MSC.232(82), which have provided performance standards for electronic chart display and information systems (ECDIS), NOTING that the up -to-date charts required by SOLAS regulations V/ 19 and V/27 can be provided and displayed electronically on board ships by ECDIS, and that the other nautical publications required by regulation V/27 may also be so provided and displayed, NOTING ALSO the recent developments and enhancement of ECDIS, inc luding new electronic navigational chart transfer functionality in the performance standards, is the --- electronic navigational chart transfer functionality in the performance standards, is the necessary step towards the implementation of the e -navigation concept of harmonized maritime services, RECOGNIZING the need to improve the revised performance standards, previously adopted by resolution MSC.232(82), for ECDIS in order to ensure the operational reliability of such equipment and taking into account the technological progress and experience gained, HAVING CONSIDERED the recommendation made by the Sub -Committee on Navigation, Communications and Search and Rescue, at its ninth session, 1 ADOPTS the revised Performance standards for electronic chart display and information systems (ECDIS), set out in the annex to the present resolution; 2 RECOMMENDS Governments to ensure that ECDIS equipment: (a) if installed on or after 1 January 2029, conforms to performance standards not inferior to those specified in the annex to the present resolution; --- not inferior to those specified in the annex to the present resolution; (b) if installed on or after 1 January 2026 but before 1 January 2029 , conforms either to performance standards not inferior to those specified in the annex to the present resolution or to performance standards not inferior to those specified in the annex to resolution MSC.232(82); RESOLUTION MSC.530(106) (adopted on 7 November 2022) PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS) MSC 106/19/Add.1 Annex 22, page 2 I:\MSC\106\MSC 106/19/Add.1.docx (c) if installed on or after 1 January 2009 but before 1 January 2026, conforms to performance standards not inferior to those specified in the annex to resolution MSC.232(82); and (d) if installed on or after 1 January 1996 but before 1 January 2009, conforms to performance standards not inferior to those specified in the annex to resolution A.817(19), as amended by resolutions MSC.64(67) and MSC.86(70); --- resolution A.817(19), as amended by resolutions MSC.64(67) and MSC.86(70); 3 AGREES that, for the purpose of this resolution, the expression installed on or after 1 January 2029 means: (a) for ships for which the building contract is placed on or after 1 January 2029, or in the absence of the contract, constructed on or after 1 January 2029, any installation date on the ship; or (b) for ships other than those prescribed in (a) above, acontractual delivery date for the equipment or, in the absence of acontractual delivery date, the actual delivery date of the equipment to the ship on or after 1 January 2029. RESOLUTION MSC.530(106) (adopted on 7 November 2022) PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS) MSC 106/19/Add.1 Annex 22, page 3 I:\MSC\106\MSC 106/19/Add.1.docx ANNEX PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS) 1 SCOPE OF ECDIS --- I:\MSC\106\MSC 106/19/Add.1.docx ANNEX PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS) 1 SCOPE OF ECDIS 1.1 The primary function of ECDIS is to contribute to safe navigation. 1.2 ECDIS with adequate backup arrangements may be accepted as complying with the up-to-date charts and nautical publications required by regulations V/19 and V/27 of the 1974 SOLAS Convention. For the purpose of this document, the definition of electronic navigational data service (ENDS) encompasses the nautical charts and nautical publications as defined in SOLAS chapter Vand IHO standards in force. 1.3 ECDIS should be capable of displaying all nautical information necessary for safe and efficient navigat ion, originated and distributed by or on the authority of agovernment , authorized hydrographic office or other relevant government institution, as required by SOLAS regulations V/19 and V/27. --- authorized hydrographic office or

other relevant government institution, as required by SOLAS regulations V/19 and V/27. 1.4 ECDIS should facilitate simple and reliable updating of the ENDS. 1.5 ECDIS should reduce the navigational workload compared to using the paper chart and paper nautical publications. It should enable the mariner to execute in aconvenient and timely manner all route planning, route monitoring and positioning. It should be capable of continuously indicating, monitoring and recording the ship'sposition. 1.6 The ECDIS display may also b eused for the display of radar, radar tracked target information, AIS and other appropriate data layers to assist in route monitoring. 1.7 ECDIS should provide appropriate alerts or indications with respect to the information displayed or malfunction of the equipment (see appendix 5). ECDIS should meet the requirements of the Performance standards for bridge alert management (resolution MSC.302(87)). --- requirements of the Performance standards for bridge alert management (resolution MSC.302(87)). 1.8 When the relevant chart information is not available in the appropriate form (see section 4), some ECDIS equipment may operate in the raster chart display system (RCDS) mode as defined in appendix 7. RCDS mode of operation should conform to performance standards not inferior to those set out in appendix 7. 2 APPLICATION OF THESE STANDARDS 2.1 These performance standards should apply to all ECDIS equipment carried on all ships, as follows: .1 dedicated stand-alone workstation; and ,2 amultifunction workstation as part of an INS. 2.2 These performance standards apply to ECDIS mode of operation, ECDIS in RCDS mode of operation as specified in appendix 7 and ECDIS backup arrangements as specified in appendix 6. RESOLUTION MSC.530(106) (adopted on 7 November 2022) PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS) --- RESOLUTION MSC.530(106) (adopted on 7 November 2022) PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS) MSC 106/19/Add.1 Annex 22, page 4 I:\MSC\106\MSC 106/19/Add.1.docx 2.3 Requirements for structure, format, encryption presentation of the ENDS are within the scope of relevant IHO standards, including those listed in appendix 1. 2.4 In addition to the general requirements set out in resolution A.694(17) 1 and the presentation requirements set out in resolution MSC.191(79), as amended, ECDIS equipment should meet the requirements of these standards and follow the relevant guidelines on ergonomic principles adopted by the Organization.2 3 DEFINITIONS For the purpose of these performance standards: 3.1 Electronic chart display and information system (ECDIS) means anavigation information system which with adequate backup arrangements can be accepted as complying --- information system which with adequate backup arrangements can be accepted as complying with the up -to-date nautical chart and nautical publications required by SOLAS regulations V/19 and V/27, by displaying selected information from asystem database with positional information from navigation sensors to assist the mariner in route planning and route monitoring and, if required, display additional navigation-related information. 3.2 Electronic navigational chart (ENC) means the database, standardized as to content, structure and format, issued for use with ECDIS by or on the authority of agovernment, authorized hydrographic office or other relevant government institution, and conforming to IHO standards. The ENC contains all the nautical chart information necessary for safe navigation. 3.3 Electronic navigational data service (ENDS) means aspecial -purpose database compiled from nautical chart and nautical publication data, standardized as to content, --- compiled from nautical chart and nautical publication data, standardized as to content, structure and format, issued for use with ECDIS by or on the authority of agovernment, authorized hydrographic office or other relevan tgovernment institution, and conforming to IHO standards; and, which is designed to meet the requirement of marine navigation and the nautical charts and nautical publications carriage requirements in SOLAS regulations V/19 and V/27. The navigational base layer of ENDS is the electronic navigational chart (ENC). 3.4 System database means adatabase, in the manufacturer'sinternal ECDIS format, resulting from the lossless transformation of the ENDS contents and its updates. It is this database that i saccessed by ECDIS for the display generation and other navigational functions, and is equivalent to up-to-date ENDS. 3.5 Standard display is the display mode intended to be used as aminimum during route --- functions, and is equivalent to up-to-date ENDS. 3.5 Standard display is the display mode intended to be used as aminimum during route planning and route monitoring. The chart content is listed in appendix 2. 3.6 Display base means the chart content as listed in appendix 2 and which cannot be removed from the display. It is not intended to be sufficient for safe navigation. 3.7 Further information on ECDIS definitions may be found in IHO Hydrographic Dictionary Publication S-32 (see appendix 1). 1 MSC/Circ.982. 2 Refer to Publication IEC 60945. RESOLUTION MSC.530(106) (adopted on 7 November 2022) PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS) MSC 106/19/Add.1 Annex 22, page 5 I:\MSC\106\MSC 106/19/Add.1.docx MODULE A – DATABASE 4

PROVISION AND UPDATING 4.1 The ENDS information to be used in ECDIS should be issued by or on the authority --- MODULE A – DATABASE 4 PROVISION AND UPDATING 4.1 The ENDS information to be used in ECDIS should be issued by or on the authority of agovernment, government -authorized hydrographic office or other relevant government institution, and conform to IHO standards as listed in appendix 1. 4.2 The contents of the system database should be adequate and up to date for the intended voyage to comply with SOLAS regulations V/19 and V/27. 4.3 It should not be possible to alter the contents of the ENDS or system database information transformed from the ENDS. The display of the content of ENDS should be compliant with IHO standards including rules set for interoperability. 4.4 ECDIS should be capable of accepting official updates to the ENDS provided in conformity with IHO standards. These updates should be automatically applied to the system database. By whatever means updates are received, the implementation procedure should not interfere with the display in use. --- database. By whatever means updates are received, the implementation procedure should not interfere with the display in use. 4.5 ECDIS should also be capable of accepting updates to the ENDS data entered manually with simple means for verification prior to the final acceptance of the data. They should be distinguishable on the display from ENDS information and its official updates and not affect display legibility. 4.6 ECDIS should keep and display on demand arecord of updates including time of application to the system database. This record should include updates for each ENDS until it is superseded by anew edition. 4.7 ECDIS should allow the mariner to display updates in order to review their contents and to ascertain that they have been included in the system database. 4.8 ECDIS should be capable of accepting ENDS in accordance with the I HO Data Protection Scheme.3 MODULE B – OPERATIONAL AND FUNCTIONAL REQUIREMENTS --- Protection Scheme.3 MODULE B – OPERATIONAL AND FUNCTIONAL REQUIREMENTS 5 DISPLAY OF SYSTEM DATABASE INFORMATION 5.1 An ECDIS should be capable of accepting and converting an ENDS and their updates into asystem database. ECDIS should be capable of displaying and processing all system database information as specified by IHO. The ECDIS may also be capable o faccepting asystem database resulting from conversion ashore, in accordance with IHO resolutions.4 5.2 System database information available for display during route planning and route monitoring should be subdivided into the following three categories: display base, standard display and all other information (see appendix 2). 5.3 ECDIS should present the standard display at any time by asingle operator action. 3 IHO Publication S -63 – Data Protection Scheme (for S -57 ENCs) and S -100, Part 15 – Data Protection Scheme (for S-100 products) (see appendix 1). --- Scheme (for S-100 products) (see appendix 1). 4 IHO Publication M-3 – Resolutions of the IHO. RESOLUTION MSC.530(106) (adopted on 7 November 2022) PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS) MSC 106/19/Add.1 Annex 22, page 6 I:\MSC\106\MSC 106/19/Add.1.docx 5.4 When an ECDIS is switched on following aswitch off or power failure, it should return to the most recent manually selected settings for display. 5.5 It should be easy to add or remove information from the ECDIS display. It should not be possible to remove information contained in the display base. 5.6 For any operator -identified geographical position (e.g. by cursor picking), ECDIS should display on demand the information about the chart obje cts associated with such aposition. 5.7 It should be possible to change the display scale by appropriate steps, e.g. by means of either chart scale values or ranges in nautical miles. --- of either chart scale values or ranges in nautical miles. 5.8 It should be possible for the mariner to select asafety contour from the information provided by the system database. ECDIS should emphasize the safety contour over other contours on the display. However: .1 if the mariner does not specify asafety contour, this should default to 30 m. If the safety contour specified by the mariner or the default 30 mcontour is not in the displayed system database, the safety contour shown should default to the next deeper contour; .2 if the safety contour in use becomes unavailable due to achange in source data, the safety contour should default to the next deeper contour; .3 in each of the above cases, an indication should be provided; and .4 the mariner should be able to select apermanent display of safety contour and safety depth settings. 5.9 It should be possible for the mariner to select asafety depth. ECDIS should --- and safety depth settings. 5.9 It should be possible for the mariner to select asafety depth. ECDIS should emphasize soundings equal to or less than the safety depth whenever spot soundings are selected for display. 5.10 It should be possible to use dynamic water level adjustment and an indication should be provided. 5.11 The ENDS and all updates to it should be displayed without any degradation of their information content. 5.12 ECDIS should provide ameans to ensure that the ENDS and all updates to it have been correctly loaded into the system database. 5.13 The ENDS data and updates to it should be clearly distinguishable from other displayed information, including those

listed in appendix 3. 6 SCALE 6.1 ECDIS should provide an indication if: .1 the information is displayed at alarger scale than that contained in the ENC; .2 own ship'sposition is covered by an ENC at alarger scale than that provided by the display; or --- .2 own ship'sposition is covered by an ENC at alarger scale than that provided by the display; or .3 information at own ship'sposition is not displayed because of applying scale minimum for display. RESOLUTION MSC.530(106) (adopted on 7 November 2022) PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS) MSC 106/19/Add.1 Annex 22, page 7 I:\MSC\106\MSC 106/19/Add.1.docx 7 DISPLAY OF OTHER NAVIGATIONAL INFORMATION 7.1 Radar information and/or AIS information may be transferred from systems compliant with the relevant standards of the Organization. Other navigational information may be added to the ECDIS display. However, it should not degrade the displayed system database information and it should be clearly distinguishable from the system database information. 7.2 It should be possible to remove the radar information, AIS information and other navigational information by single operator action. --- 7.2 It should be possible to remove the radar information, AIS information and other navigational information by single operator action. 7.3 ECDIS and added navigational information should use acommon reference system. If this is not the case, an indication should be provided. 7.4 Radar 7.4.1 Transferred radar information may contain aradar image and/or tracked target information. 7.4.2 If the radar image is added to the ECDIS display, the chart and the radar image should match in scale, projection and orientation. 7.4.3 The radar image and the position from the position sensor should both be adjusted automatically for antenna offset from the conning position. 8 DISPLAY MODE AND GENERATION OF THE NEIGHBOURING AREA 8.1 It should always be possible to display the system database information in a "north-up" orientation. Other orientations are permitted. When such orientations are displayed, --- "north-up" orientation. Other orientations are permitted. When such orientations are displayed, the orientation should be altered in steps large enough to avoid unstable display of the chart information. 8.2 ECDIS should provide for true motion mode. Other modes are permitted. 8.3 When true motion mode is in use, reset and generation of the chart display of the neighbouring area should take place automatically at own ship'sdistance from the edge of the display as determined by the mariner. 8.4 It should be possible to manually change the displayed chart area and the position of own ship relative to the edge of the display. 8.5 If the area covered by the ECDIS display includes waters for which no ENC at ascale appropriate for navigation is available, the areas representing those waters should carry an indication (see appendix 5) to the mariner to refer to the paper chart or to the RCDS mode of operation (see appendix 7). 9 COLOURS AND SYMBOLS --- operation (see appendix 7). 9 COLOURS AND SYMBOLS 9.1 IHO-recommended colours and symbols should be used to represent system database information.5 5 IHO Publication S-52 - Specifications for Chart Content and Display Aspects of ECDIS and S-101 – Portrayal Catalogue (see appendix 1) and S-98. RESOLUTION MSC.530(106) (adopted on 7 November 2022) PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS) MSC 106/19/Add.1 Annex 22, page 8 I:\MSC\106\MSC 106/19/Add.1.docx 9.2 The colours and symbols other than those mentioned in 9.1 should comply with the applicable requirements contained in the IMO standards for navigational symbols.6 9.3 ECDIS should allow the mariner to select whether own ship is displayed in true scale or as asymbol. 10 DISPLAY REQUIREMENTS 10.1 ECDIS should be capable of displaying information for: .1 route planning and supplementary navigation tasks; and .2 route monitoring. --- 10.1 ECDIS should be capable of displaying information for: .1 route planning and supplementary navigation tasks; and .2 route monitoring. 10.2 The effective size of the chart presentation for route monitoring should be at least 270 mm x 270 mm. 10.3 The display should be capable of meeting the colour and resolution recommendations of IHO.5 10.4 The method of presentation should ensure that the displayed information is clearly visible to more than one observer in the conditions of light normally experienced on the bridge of the ship by day and by night. 10.5 If information categories included in the standard display (see appendix 2) are removed to customize the display, this should be permanently indicated. Identification of categories which are removed from the standard display should be shown on demand. 11 ROUTE PLANNING, MONITORING AND VOYAGE RECORDING 11.1 It should be possible to carry out route planning, route monitoring in asimple and --- 11 ROUTE PLANNING, MONITORING AND VOYAGE RECORDING 11.1 It should be possible to carry out route planning, route monitoring in asimple and reliable manner. 11.2 The largest scale data available in the system database for the area given should always be used by the ECDIS for all alerts or indications of crossing the ship'ssafety contour and of entering aprohibited area, and for alerts and indications according to appendix 5. 11.3 Route planning 11.3.1 It should

be possible to carry out route planning including both straight and curved segments. 11.3.2 It should be possible to adjust aplanned route alphanumerically and graphically including: .1 adding waypoints to aroute; .2 deleting waypoints from aroute; and .3 changing the position of awaypoint. 6 SN.1/Circ.243/Rev.2. 5 IHO Publication S-52 - Specifications for Chart Content and Display Aspects of ECDIS and S-101 – Portrayal Catalogue (see appendix 1) and S-98. --- Catalogue (see appendix 1) and S-98. RESOLUTION MSC.530(106) (adopted on 7 November 2022) PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS) MSC 106/19/Add.1 Annex 22, page 9 I:\MSC\106\MSC 106/19/Add.1.docx 11.3.3 It should be possible to plan one or more alternative routes in addition to the selected route. The selected route should be clearly distinguishable from the other routes. 11.3.4 Agraphical indication is required if the mariner plans aroute closer than auser-specified distance from own ship'ssafety contour. 11.3.5 Agraphical indication should be given if the mariner plans aroute closer than auser-specified distance from the boundary of auser-selectable category of prohibited area or geographic area for which special conditions exist (see appendix 4). Agraphic al indication should also be given if the mariner plans aroute closer than auser -specified distance from a --- should also be given if the mariner plans aroute closer than auser -specified distance from auser-selectable category of point objects, such as afixed or floating aid to navigation or isolated danger. The user -selectable categories should be the same as the user selections for the display of objects and be based on IHO standards. There should be apermanent indication when any user -selectable categories are deselected. Details of the deselection should be available on demand. 11.3.6 It should be possible for the mariner to select that the indications of 11.3.4 and 11.3.5 take into account accuracy information of relevant hydrographic information, as defined by IHO standards. 11.3.7 It should also be possible to perform acomplete route check to support the appraisal and planning process according to the applicable parts of resolution A.893(21). Detected objects should be available for review in graphical form and, on demand, in textual form. --- objects should be available for review in graphical form and, on demand, in textual form. 11.3.8 It should be possible for the mariner to specify across track limit of deviation from the planned route at which an automatic off-track alarm should be activated. 11.4 Route monitoring 11.4.1 For route monitoring the selected route and own ship'sposition should appear whenever the display covers that area. 11.4.2 It should be possible to display asea area that does not have the ship on the display (e.g. for look ahead, route planning), while route monitoring. If this is done on the display used for route monitoring, the automatic route monit oring functions (e.g. updating ship'sposition, and providing alerts and indications) should be continuous. It should be possible to return to the route monitoring display covering own ship'sposition immediately by single operator action. 11.4.3 It should be possible to select that ECDIS gives an alarm and related graphical --- 11.4.3 It should be possible to select that ECDIS gives an alarm and related graphical indication if, within aspecified time or distance set by the mariner, own ship will pass closer than auser-selected distance from the safety contour. There should be apermanent indication when the safety contour alarm is deselected. 11.4.4 ECDIS should give awarning or caution, or indication, as selected by the mariner, and related graphical indication if, within aspecified time or distance set by the mariner, own ship will pass closer than auser -selected distance from the boundary of auser -selectable category of prohibited area or of ageographical area for which special conditions exist (see appendix 4). The user -selectable categories should be the same as user selec tions for the display of objects and be based on IHO standards. There should be apermanent indication when any user -selectable categories are deselected. Details of the deselection should be --- when any user -selectable categories are deselected. Details of the deselection should be available on demand. 11.4.5 An alarm should be given when the specified cross track limit for deviation from the selected route, if defined by the mariner when route planning, is exceeded. RESOLUTION MSC.530(106) (adopted on 7 November 2022) PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS) MSC 106/19/Add.1 Annex 22, page 10 I:\MSC\106\MSC 106/19/Add.1.docx 11.4.6 ECDIS should give awarning or caution or indication as selected by the mariner and related graphical indication if, continuing on its present course and speed, over aspecified time or distance set by the mariner, own ship will pass closer than auser -specified distance from auser-selectable category of danger (e.g. obstruction, wreck, rock) that is shallower than the mariner'ssafety contour or auser -selectable category of aid to navigation. The --- the mariner'ssafety contour or auser -selectable category of aid to navigation. The user-selectable categories should be the same as user selections for the display of objects and be based on IHO standards. There should be apermanent indicatio nwhen any of

the user-selectable categories are deselected. Details of the deselection should be available on demand. 11.4.7 Agraphical indication should be given if the current or the next leg of the selected route passes closer than auser-specified distance from the safety contour. 11.4.8 Agraphical indication should be given if the current or the next leg of the selected route goes closer than auser -specified distance from the boundary of auser -selectable category of prohibited ar ea or ageographic area for which special conditions exist (see appendix 4). Agraphical indication should also be given if the selected route goes closer than --- appendix 4). Agraphical indication should also be given if the selected route goes closer than auser-specified distance from auser-selectable category of point objects, such as afixed or floating aid to navigation or isolated danger. The user -selectable categories should be the same as user selections for the display of objects and be based on IHO standards. 11.4.9 It should be possible for the mariner to select that the indications of 11. 4.3, 11.4.4, 11.4.6, 11.4.7 and 11.4.8 take into account accuracy information of relevant hydrographic information, as defined by IHO standards. 11.4.10 The ship'sposition should be derived from acontinuous positioning system of an accuracy consistent with the requirements of safe navigation. Whenever possible, asecond independent positioning source, preferably of adifferent type, should be provided. In such cases, ECDIS should be capable of identifying discrepancies between the two sources. --- cases, ECDIS should be capable of identifying discrepancies between the two sources. 11.4.11 ECDIS should provide awarning when the input from position, heading or speed sources is lost. ECDIS should also repeat, but only as an indication, any alerts or indication passed to it from position, heading or speed sources. 11.4.12 Awarning should be given by ECDIS when the ship reaches aspecified time or distance, set by the mariner, in advance of acritical point on the planned route. 11.4.13 The positioning system and the system database should be on the same geodetic datum. ECDIS should give awarning if this is not the case. 11.4.14 It should be possible to display alternative routes in addition to the selected route. The selected route should be clearly distinguishable from the other routes. During the voyage, it should be possible for the mariner to modify the selected route or change to an alternative route. 11.4.15 It should be possible to display: --- should be possible for the mariner to modify the selected route or change to an alternative route. 11.4.15 It should be possible to display: .1 time labels along aship'strack manually on demand and automatically at intervals selected between 1 and 120 minutes; and .2 an adequate number of points, free movable electronic bearing lines, variable and fixed range markers and other symbols required for navigation purposes and specified in appendix 3.

 11.4.16 It should be possible to enter the geographical coordinates of any position and then display that position on demand. Also, it should be possible to select any point (features, symbol or position) on the display and read its geographical coordinates on demand. --- symbol or position) on the display and read its geographical coordinates on demand. 11.4.17 It should be possible to adjust the displayed geographic position of the ship manually. This manual adjustment should be indicated alphanumerically on the screen, maintained until altered by the mariner and automatically recorded. 11.4.18 ECDIS should provide the capability to enter and plot manually obtained bearing and distance lines of position (LOP), and calculate the resulting position of own ship. It should be possible to use the resulting position as an origin for dead reckoning. 11.4.19 ECDIS should indicate discrepancies between the positions obtained by continuous positioning systems and positions obtained by manual observations. 11.5 Voyage recording 11.5.1 ECDIS should store and be able to reproduce certain minimum elements required to reconstruct the navigation and verify the official database used during the previous 12 hours. --- reconstruct the navigation and verify the official database used during the previous 12 hours. The following data should be recorded at one minute intervals: .1 to ensure arecord of own ship'spast track: time, position, heading, and speed; .2 to ensure arecord of official data used: ENC source, edition, date, cell and update history.; and .3 any changes in safety contour, look ahead and route monitoring alert settings. 11.5.2 ECDIS should output the information listed in 11.5.1.2 and 11.5.1.3 to avoyage data recorder 11.5.3 In addition, ECDIS should record the complete track for the entire voyage, with time marks at intervals not exceeding 4 hours. 11.5.4 It should not be possible to manipulate or change the recorded information. 11.5.5 ECDIS should have acapability to preserve the record of the previous 12 hours and the voyage track. 12 CALCULATIONS AND ACCURACY --- 11.5.5 ECDIS should have acapability to preserve the record of the previous 12 hours and the voyage track. 12 CALCULATIONS AND ACCURACY 12.1 The accuracy of all calculations performed by ECDIS

should be independent of the characteristics of the output device and should be consistent with the system database accuracy. 12.2 Bearings and distances drawn on the display or those measured between features already drawn on the display should have accuracy no less than that afforded by the resolution of the display. RESOLUTION MSC.530(106) (adopted on 7 November 2022) PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS) MSC 106/19/Add.1 Annex 22, page 12 I:\MSC\106\MSC 106/19/Add.1.docx 12.3 The system should be capable of performing and presenting the results of at least the following calculations: .1 true distance and azimuth between two geographical positions; --- following calculations: .1 true distance and azimuth between two geographical positions; .2 geographic position from known position and distance/azimuth; and .3 geodetic calculations such as spheroidal distance, rhumb line and great circle. 13 PERFORMANCE TESTS, MALFUNCTIONS ALERTS AND INDICATIONS 13.1 ECDIS should be provided with means for either automatically or manually carrying out onboard tests of major functions. In case of afailure, the test should display information to indicate which module is at fault. 13.2 ECDIS should provide asuitable warning or indication of system malfunction. 14 BACKUP ARRANGEMENTS Adequate backup arrangements should be provided to ensure safe navigation in case of an ECDIS failure; see appendix 6. .1 Facilities enabling asafe takeover of the ECDIS functions should be provided in order to ensure that an ECDIS failure does not develop into acritical situation. --- provided in order to ensure that an ECDIS failure does not develop into acritical situation. .2 Abackup arrangement should provide means of safe navigation for the remaining part of avoyage in the case of an ECDIS failure. MODULE C – INTERFACING AND INTEGRATION 15 CONNECTIONS WITH OTHER EQUIPMENT7 15.1 ECDIS should not degrade the performance of any equipment providing sensor inputs. Nor should the connection of optional equipment degrade the performance of ECDIS below this standard. 15.2 ECDIS should be connected to the ship'sposition fixing system, to the gyro compass and the speed and distance measuring device. For ships not fitted with agyro compass, ECDIS should be connected to amarine transmitting heading device. 15.3 ECDIS may provide ameans to supply system database information to external equipment. 16 POWER SUPPLY 16.1 It should be possible to operate ECDIS and all equipment necessary for its normal --- equipment. 16 POWER SUPPLY 16.1 It should be possible to operate ECDIS and all equipment necessary for its normal functioning when supplied by an emergency source of electrical power in accordance with the appropriate requirements of SOLAS chapter II-1. 16.2 Changing from one source of power supply to another or any interruption of the supply for aperiod of up to 45 seconds should not require the equipment to be manually re-initialized. 7 Publication IEC 61162. RESOLUTION MSC.530(106) (adopted on 7 November 2022) PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS) MSC 106/19/Add.1 Annex 22, page 13 I:\MSC\106\MSC 106/19/Add.1.docx APPENDIX 1 REFERENCE DOCUMENTS The following international organizations have developed technical standards and specifications, as listed below, for use in conjunction with this standard. The latest edition of these documents should be obtained from the organization concerned: --- these documents should be obtained from the organization concerned: INTERNATIONAL MARITIME ORGANIZATION (IMO) Address: International Maritime Organization Phone: +44 207 735 76 11 4 Albert Embankment Fax: +44 207 587 32 10 London SE1 7SR Email: info@imo.org United Kingdom Web: Publications Resolution MSC.191(79), as amended by resolution MSC.466(101), on Performance standards for the presentation of navigation -related information on shipborne navigational displays Resolution A.694(17) on Recommendations on general requirements for shipborne radio equipment forming part of the Global Maritime Distress and Safety System (GMDSS) and for electronic navigational aids Resolution MSC.302(87) on Performance standards for bridge alert management SN.1/Circ.207 on Differences between RCDS and ECDIS SN.1/Circ.243/Rev.2 on Guidelines fo rthe presentation of navigation -related symbols, terms and abbreviations --- SN.1/Circ.243/Rev.2 on Guidelines fo rthe presentation of navigation -related symbols, terms and abbreviations MSC/Circ.982 on Guidelines on ergonomic criteria for bridge equipment and layout INTERNATIONAL HYDROGRAPHIC ORGANIZATION (IHO) Address: Directing Committee Phone: +377 93 10 81 00 International Hydrographic Organization BP 445 Fax: +377 93 10 81 40 MC 98011 Monaco Cedex Email: info@iho.int Principality of Monaco Web: Publications IHO Publication S -52, Specifications for Chart Content and Display Aspects of ECDIS IHO Publication S-52 appendix 1, Guidance on Updating the Electronic Navigational Chart IHO Publication S-52 appendix 2, Colour and Symbol Specifications for ECDIS IHO Publication S-32, Hydrographic Dictionary RESOLUTION MSC.530(106) (adopted on 7 November 2022) PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS) MSC 106/19/Add.1 Annex 22, page 14 --- PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS) MSC 106/19/Add.1 Annex 22, page

14 I:\MSC\106\MSC 106/19/Add.1.docx IHO Publication S-57, IHO Transfer Standard for Digital Hydrographic Data IHO Publication S-100, IHO Universal Hydrographic Data Model IHO Publication S-101 – ENC Product Specification IHO Publication S-98 – Data Product Interoperability in S-100 Navigation Systems IHO Publication S-61, IHO Product specification for Raster Navigational Charts (RNC) IHO Publication S-63, IHO Data Protection Scheme IHO Publication M-3, Resolutions of the IHO INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC) Address: IEC Central Office Phone: +41 22 734 01 /50 3 rue de Varembe Fax: +41 22 733 38 43 PO Box 131 CH-1211 Geneva 20 Switzerland Publications IEC Publication 61174, Electronic Chart Display and Information Systems (ECDIS) - --- PO Box 131 CH-1211 Geneva 20 Switzerland Publications IEC Publication 61174, Electronic Chart Display and Information Systems (ECDIS) - Operational and Performance Requirements, Method of Testing and Required Test Results. IEC Publication 60945, General Requirements for Shipborne Radio Equipmen t Forming Part of the Global Maritime Distress and Safety System and Marine Navigational Equipment. IEC Publication 61162, Digital lnterfaces – Navigation and Radiocommunication Equipment On board Ship. IEC Publication 62288, Maritime Navigation and Radi ocommunication Equipment and Systems – Presentation of navigation -related information – General requirements, methods of test and required test results. RESOLUTION MSC.530(106) (adopted on 7 November 2022) PERFORMANCE STANDARDS FOR ELECTRONIC CHART DISPLAY AND INFORMATION SYSTEMS (ECDIS) --- RESOLUTION MSC.533(107) (adopted on 8 June 2023) AMENDMENTS TO THE PROTOCOL OF 1978 RELATING TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974 RESOLUTION MSC.533(107) (adopted on 8 June 2023) AMENDMENTS TO THE PROTOCOL OF 1978 RELATING TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974 THE MARITIME SAFETY COMMITTEE, RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee, RECALLING ALSO article VIII(b) of the International Convention for the Safety of Life at Sea, 1974 ("the Convention") and article II of the Protocol of 1978 relating to the Convention ("the 1978 SOLAS Protocol") concerning the procedure for amending the 1978 SOLAS Protocol, HAVING CONSIDERED, at its 107th session, amendments to the 1978 SOLAS Protocol propose